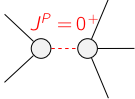
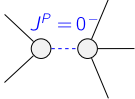
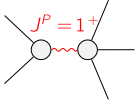
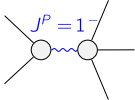


| $\mathcal{K}_0^{\#1}$                           |                                                              | $\mathcal{K}_0^{\#1}{}_{\alpha\beta\chi}$                    |                                                                         |                                                                         |                                                 |                                           |   |
|-------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------|---|
| $\mathcal{K}_0^{\#1} \dagger$                   | $\frac{1}{2}(-3\Upsilon_1 k^2 - 3\overset{(2)}{\kappa}_3)$   | 0                                                            |                                                                         |                                                                         |                                                 |                                           |   |
| $\mathcal{K}_0^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                                            | $3k^2\overset{(4)}{\kappa}_{15} + 3\overset{(2)}{\kappa}_1$  | $\mathcal{K}_1^{\#1}{}_{\alpha\beta}$                                   | $\mathcal{K}_1^{\#2}{}_{\alpha\beta}$                                   | $\mathcal{K}_1^{\#1}{}_{\alpha}$                | $\mathcal{K}_1^{\#2}{}_{\alpha}$          |   |
| $\mathcal{K}_1^{\#1} \dagger^{\alpha\beta}$     | $\Upsilon_2 k^2 + 2\overset{(2)}{\kappa}_1$                  | $\frac{\Upsilon_2 k^2 + 2\overset{(2)}{\kappa}_1}{\sqrt{2}}$ | 0                                                                       | 0                                                                       |                                                 |                                           |   |
| $\mathcal{K}_1^{\#2} \dagger^{\alpha\beta}$     | $\frac{\Upsilon_2 k^2 + 2\overset{(2)}{\kappa}_1}{\sqrt{2}}$ | $\frac{1}{2}(\Upsilon_2 k^2 + 2\overset{(2)}{\kappa}_1)$     | 0                                                                       | 0                                                                       |                                                 |                                           |   |
| $\mathcal{K}_1^{\#1} \dagger^{\alpha}$          | 0                                                            | 0                                                            | $-k^2\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3$                 | $\frac{k^2\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$ |                                                 |                                           |   |
| $\mathcal{K}_1^{\#2} \dagger^{\alpha}$          | 0                                                            | 0                                                            | $\frac{k^2\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$ | $\frac{1}{2}(-k^2\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3)$    | $\mathcal{K}_2^{\#1}{}_{\alpha\beta}$           | $\mathcal{K}_2^{\#1}{}_{\alpha\beta\chi}$ |   |
|                                                 |                                                              |                                                              |                                                                         |                                                                         | $\mathcal{K}_2^{\#1} \dagger^{\alpha\beta}$     | 0                                         | 0 |
|                                                 |                                                              |                                                              |                                                                         |                                                                         | $\mathcal{K}_2^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                         | 0 |

|                                                 |                                             | $\mathcal{J}_{0^+}^{\#1}$           | $\mathcal{J}_{0^+}^{\#1\alpha\beta\chi}$                             |                                                 |                                      |                                      |                                          |
|-------------------------------------------------|---------------------------------------------|-------------------------------------|----------------------------------------------------------------------|-------------------------------------------------|--------------------------------------|--------------------------------------|------------------------------------------|
| $\mathcal{J}_{0^+}^{\#1\dagger}$                |                                             | $\frac{1}{\text{Det}(0^+)}$         | 0                                                                    |                                                 |                                      |                                      |                                          |
| $\mathcal{J}_{0^+}^{\#1\dagger\alpha\beta\chi}$ |                                             | 0                                   | $\frac{1}{3(k^2\overset{(4)}{\kappa}_{15}+\overset{(2)}{\kappa}_1)}$ | $\mathcal{J}_{1^+}^{\#1\alpha\beta}$            | $\mathcal{J}_{1^+}^{\#2\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#1\alpha}$      | $\mathcal{J}_{1^+}^{\#2\alpha}$          |
|                                                 | $\mathcal{J}_{1^+}^{\#1\dagger\alpha\beta}$ | $\frac{2}{3\text{Det}(1^+)}$        | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$                                  | 0                                               | 0                                    |                                      |                                          |
|                                                 | $\mathcal{J}_{1^+}^{\#2\dagger\alpha\beta}$ | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$ | $\frac{1}{3\text{Det}(1^+)}$                                         | 0                                               | 0                                    |                                      |                                          |
|                                                 | $\mathcal{J}_{1^+}^{\#1\dagger\alpha}$      | 0                                   | 0                                                                    | $\frac{2}{3\text{Det}(1^+)}$                    | $-\frac{\sqrt{2}}{3\text{Det}(1^+)}$ |                                      |                                          |
|                                                 | $\mathcal{J}_{1^+}^{\#2\dagger\alpha}$      | 0                                   | 0                                                                    | $-\frac{\sqrt{2}}{3\text{Det}(1^+)}$            | $\frac{1}{3\text{Det}(1^+)}$         | $\mathcal{J}_{2^+}^{\#1\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}$ |
|                                                 |                                             |                                     |                                                                      | $\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta}$     | 0                                    | 0                                    |                                          |
|                                                 |                                             |                                     |                                                                      | $\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta\chi}$ | 0                                    | 0                                    |                                          |

Abbreviations used in matrices

|                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2$ && $\Upsilon_2 == 2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3$ &&                                                                                                |
| $\text{Det}(0^+) == -\frac{3}{2}(k^2(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) + \overset{(2)}{\kappa}_3)$ && $\text{Det}(1^+) == k^2(3\overset{(4)}{\kappa}_{15} - \frac{3\overset{(4)}{\kappa}_3}{2}) + 3\overset{(2)}{\kappa}_1$ |

| Lagrangian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $\overset{(2)}{\kappa}_1\mathcal{K}_{\alpha\beta\chi}\mathcal{K}^{\alpha\beta\chi}-2\overset{(2)}{\kappa}_1\mathcal{K}^{\alpha\beta\chi}\mathcal{K}_{\beta\alpha\chi}+\overset{(2)}{\kappa}_3\mathcal{K}^{\alpha\beta}\mathcal{K}^{\chi}_{\beta\chi}+$ $\overset{(4)}{\kappa}_1\partial_\beta\mathcal{K}^{\delta}_{\chi\delta}\partial^{\chi}\mathcal{K}^{\alpha\beta}_{\alpha}+\overset{(4)}{\kappa}_2\partial_\chi\mathcal{K}^{\delta}_{\beta\delta}\partial^{\chi}\mathcal{K}^{\alpha\beta}_{\alpha}+\overset{(4)}{\kappa}_3\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}_{\alpha\chi}-2\overset{(4)}{\kappa}_3\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}_{\beta\chi}-$ $\overset{(4)}{\kappa}_3\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}_{\chi\alpha}-\frac{1}{2}\overset{(4)}{\kappa}_3\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^{\delta}_{\beta\chi}+\overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi}-2\overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\beta\alpha\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi}$ |                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                  |
| Added source term(s):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\mathcal{K}^{\alpha\beta\chi}\mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                  |
| Source constraint(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | # constraint(s)                                              | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                  |
| $\mathcal{J}_{1^+}^{\#1\alpha}+2\mathcal{J}_{1^+}^{\#2\alpha}==0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                            | $\partial_\chi\partial^\alpha\mathcal{J}^{\beta\chi}_{\beta}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}_{\beta}==3\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                  |
| $\mathcal{J}_{1^+}^{\#1\alpha\beta}==\mathcal{J}_{1^+}^{\#2\alpha\beta}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                            | $3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}+2\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta}==3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                  |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}==0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5                                                            | $3\partial_\epsilon\partial_\delta\partial^{\chi}\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\partial_\epsilon\partial^\epsilon\partial^{\chi}\partial^\alpha\mathcal{J}^{\delta\beta}_{\delta}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^{\chi}\mathcal{J}^{\beta\alpha\delta}+$ $4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^{\chi}\mathcal{J}^{\delta\alpha\beta}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}_{\delta}{}^\epsilon+3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}_{\delta}==$ $3\partial_\epsilon\partial_\delta\partial^{\chi}\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\partial_\epsilon\partial^\epsilon\partial^{\chi}\partial^\beta\mathcal{J}^{\delta\alpha}_{\delta}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^{\chi}\mathcal{J}^{\alpha\beta\delta}+$ $2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}_{\delta}{}^\epsilon+3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}_{\delta}$ |                                                                  |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta}==0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5                                                            | $3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\eta^{\alpha\beta}\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi\delta}_{\chi}{}^\delta==2\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi\delta}_{\chi}{}^\delta+3(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                  |
| Total # constraint(s):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 16                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                  |
| Resolved pole(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | # polarization(s)                                            | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Residue                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                            | $-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_{1+}\overset{(4)}{\kappa}_2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $-\frac{2}{3(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2)}$  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                            | $-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_{15}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $-\frac{1}{3\overset{(4)}{\kappa}_{15}}$                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                            | $\frac{2\overset{(2)}{\kappa}_1}{-2\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\frac{2}{6\overset{(4)}{\kappa}_{15}-3\overset{(4)}{\kappa}_3}$ |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                            | $-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{2}{3\overset{(4)}{\kappa}_2}$                             |
| Resolved unitarity condition(s):                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                              | (Demonstrably impossible)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                  |